

LM317 DC Power Supply Problems

Why not use a variable resistor for a variable DC power supply?

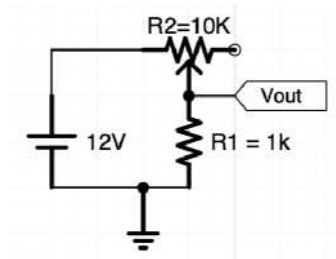
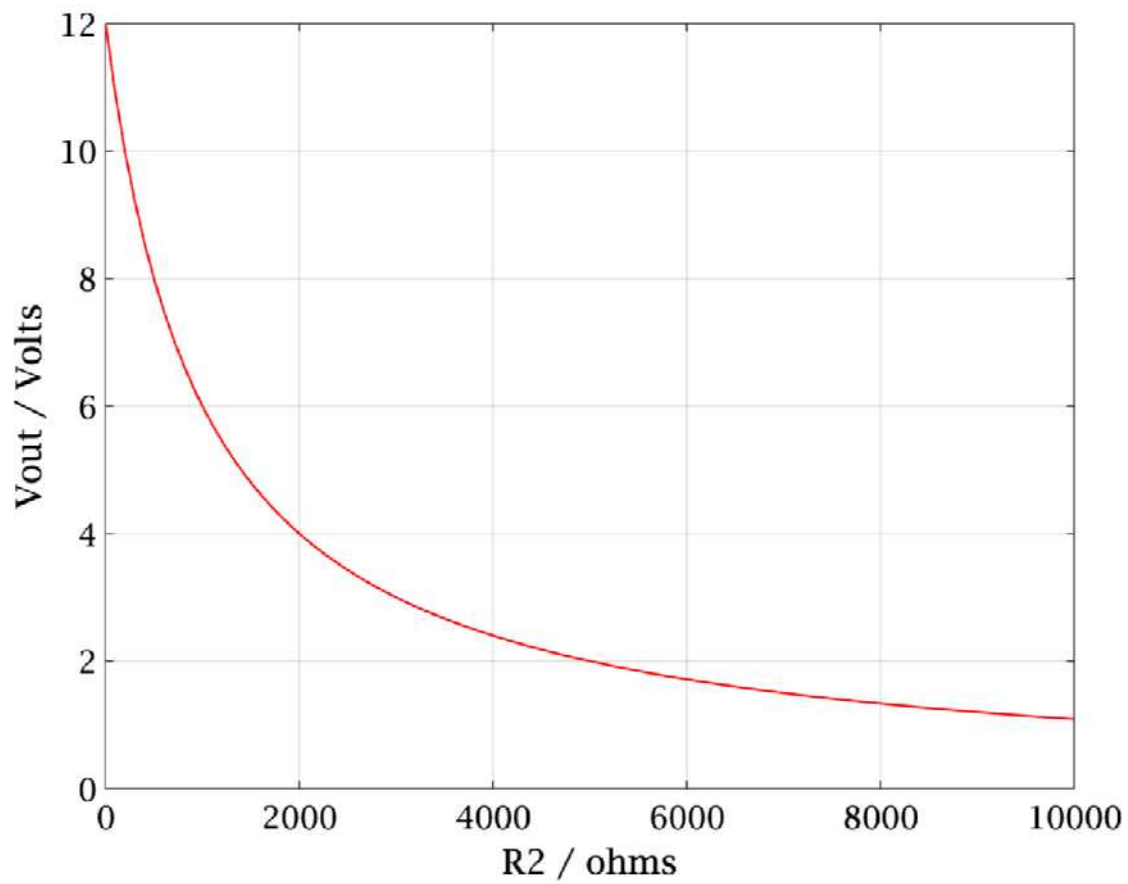


Fig. 1. Variable DC voltage source.

What's the output voltage?

```
Vin = 12;  
R1 = 1e3;  
R2 = linspace(0,1e4)';  
Vout = Vin * R1 ./ (R1+R2);  
plot(R2, Vout);  
xlabel('R2 / ohms'); ylabel('Vout / Volts')  
grid;
```



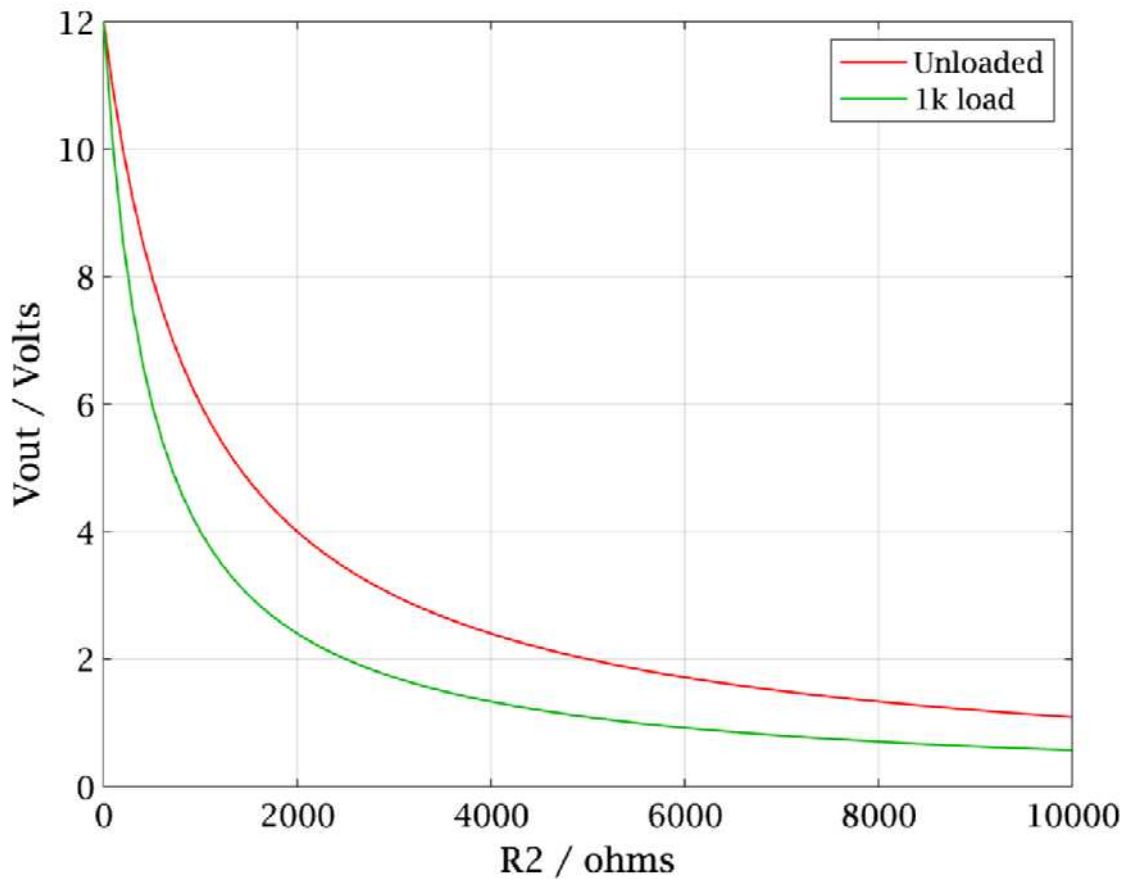
From an ergonomic perspective, the voltage changes from 12 to 4V turning R2 just 1/5 of its rotation, and the last 1/2 of rotation changes the voltage from ~2V to

```
min(Vout)
```

```
ans = 1.0909
```

That nonlinearity can make it hard to adjust, especially to large . What if we connect a load resistor, R1 to the output and draw some current? R1 is in parallel with R1, which changes Vout:

```
R1 = 1e3; % explore other values
Rparallel = R1 * R1 / (R1 + R1);
VoutP = Vin * Rparallel ./ (Rparallel+R2);
plot(R2, [Vout VoutP]);
xlabel('R2 / ohms'); ylabel('Vout / Volts')
legend('Unloaded','1k load')
grid;
```



The big change output voltage with load means the equivalent Rthevenin of this power supply is large. We can compute Rthevenin from these two curves, and we wrote a function to do this:

```
function Rth = rthevenin(Vth, V1, Rload)
Rth = (Vth - V1)*Rload / V1;
```

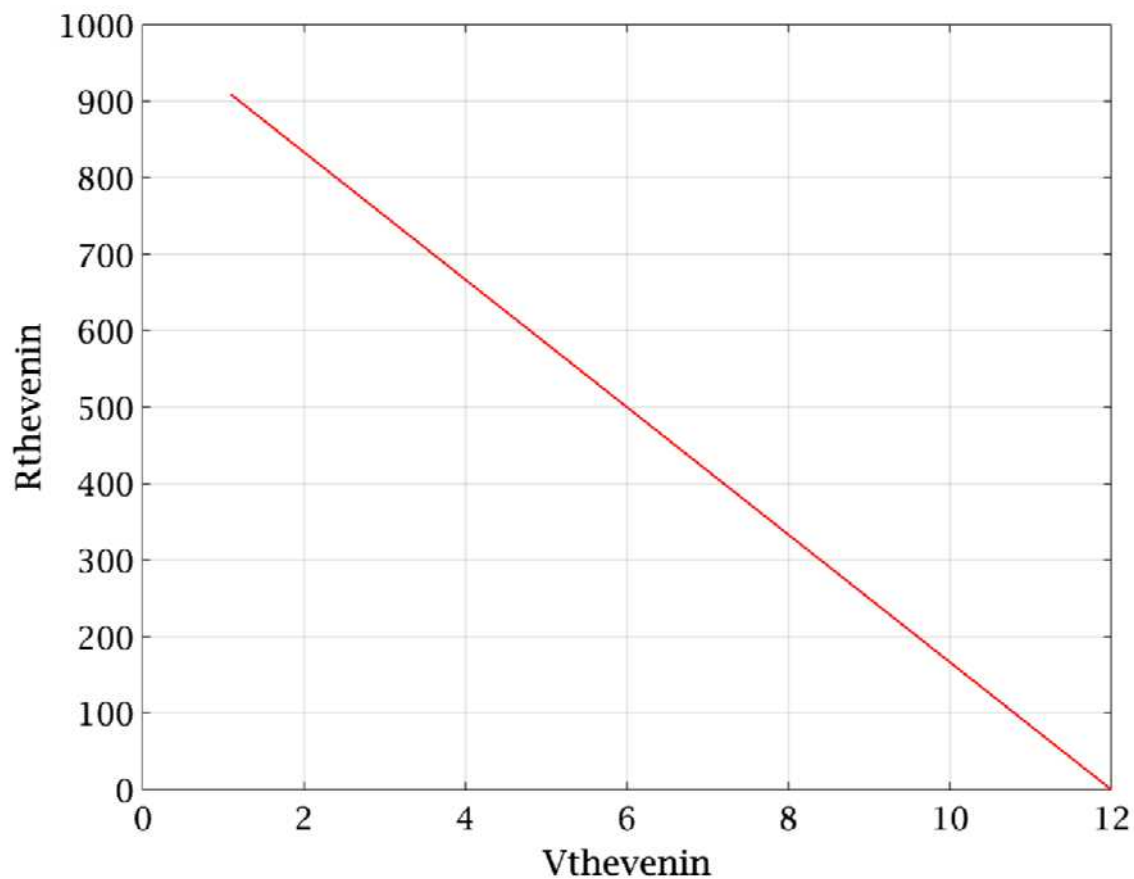
In this case, we have 100 Vth and V1 values, will rthevenin work for vectors? You can try it ... look at the size of the result.

The problem is the division of a vector by a vector has to be done "pointwise", i.e., replace / with ./:

```
function Rth = rthevenin(Vth, V1, Rload)
Rth = (Vth - V1)*Rload ./ V1;
```

Now

```
Rthevenin = rthevenin(Vout, VoutP, R1);
plot(Vout, Rthevenin)
xlabel('Vthevenin'); ylabel('Rthevenin'); grid
```



Only at 12V (where $R_2 = 0$) is R_{th} small, and thus the power supply will put out a constant voltage independent of R_{load} .

Energy usage